

Jialing Dai

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Assistant Professor | Central University of Finance and Economics
Ph.D., Computer Applied Technology | University of Chinese Academy of Sciences
Combinatorial optimization; optimization for machine learning; reinforcement learning and agents; network science and social computing

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Research interests
Combinatorial Optimization and Algorithm Theory: Submodular and non-submodular optimization; approximation and randomized algorithms.
Optimization for Machine Learning: Stochastic and nonconvex optimization; deep learning optimizers.
Reinforcement Learning and Agent Decision-Making: Graph neural networks and deep reinforcement learning.
Network Science and Social Computing: Collective behavior and opinion dynamics; rumor containment; information provenance.

APPOINTMENTS AND EDUCATION

Nov. 2026 – Present	**Central University of Finance and Economics** Assistant Professor	Beijing, China
Sep. 2020 – Oct. 2026	**University of Chinese Academy of Sciences (UCAS)** Ph.D., Computer Applied Technology; integrated master's-to-Ph.D. program. Advisor: Prof. Jianming Zhu. Doctoral stage: Mar. 2023–Oct. 2026. Dissertation: Opinion Propagation Models and Group-Aware Influence Optimization Algorithms in Social Networks.	Beijing, China
Jan. 2025 – Jan. 2026	**The University of Texas at Dallas** Research Scholar, Department of Computer Science. Faculty hosts: Profs. Weili Wu and Ding-Zhu Du.	Richardson, TX, USA
Sep. 2016 – Jun. 2020	**Beijing University of Chemical Technology (BUCT)** Bachelor of Management, Information Management and Information Systems.	Beijing, China

SELECTED RESEARCH WORKS

[1] **Jialing Dai**, Yisheng Zhou, Yefeng Sun, Jianming Zhu, Weili Wu.
"Unified Group-Aware Influence Maximization with Generalized Deep Reinforcement Learning."
IEEE Transactions on Networking (**CCF A, JCR Q1**), 34: 6198–6213, 2026. [DOI ↗]

Developed UGIP, a **unified, parameterized framework for group-aware influence propagation** that represents echo-chamber, topic-aware, homophily, and higher-order effects within a common modeling interface. Established **NP-hardness and #P-hardness** results and analyzed when group interactions invalidate submodularity and supermodularity. Designed UG-DCGNN, which combines coupled **graph neural networks with deep reinforcement learning** (Double DQN) and incorporates group evidence through input-side gated fusion to learn sequential seed-selection policies. The architecture is reusable across propagation settings, with a separate policy trained for each setting; **inference requires no Monte Carlo simulation**. Evaluated influence quality and computational efficiency on synthetic and real networks, with ablation and sensitivity studies examining group inputs, sequential re-scoring, marginal-gain rewards, signed descriptors, and alignment between group structure and network topology.

[2] **Jialing Dai**, Jianming Zhu, Guoqing Wang.
"Competitive net influence maximization on intergroup debate effect."
Information Sciences (**CAS Zone 1 (Top), JCR Q1**), 680: 121139, 2024. [DOI ↗]

Developed the Intergroup Debate Independent Cascade (IDIC) model to explain how **feedback between group consensus and individual behavior** can amplify both discussion and negative influence. Used a **group hypergraph** to couple cross-group interactions with member-level propagation, and formulated the Controversial Net Influence Maximization (CNIM) problem. Established **NP-hardness and non-submodularity**, constructed submodular upper and lower bounds, and applied a sandwich framework to obtain **data-dependent approximation guarantees**. Introduced a heterogeneous competitive cascade model with information-type-dependent activation probabilities, together with **reverse-sampling-based estimation** and the HCISG seed-selection algorithm. Improved efficiency through sampling refinement, tailored data structures, and pruning. Experiments on synthetic and real networks assessed solution quality and scalability and showed how stronger group cohesion can amplify debate-driven information diffusion.

[3] **Jialing Dai**, Jianming Zhu, Guoqing Wang.
"Opinion influence maximization problem in online social networks based on group polarization effect."
Information Sciences (**CAS Zone 1 (Top), JCR Q1**), 609: 195–214, 2022. [DOI ↗]

Built a **Zhihu dataset of 50 discussion threads and 127,178 replies** to investigate group polarization and inform the modeling of online opinion propagation. Proposed the Enhanced Independent Cascade (EIC) model, combining **individual preference updates**, positive and negative activation, and group-level polarization. The model captures how exposure to persuasive information can reinforce agreement or provoke an opposing response when preferences differ substantially. Developed GEIM, a **group-based seed-selection algorithm** that combines structure-aware candidate screening, local influence estimation, and iterative seed adjustment. Analyzed computational and structural properties and evaluated how initial preferences, group size, and the surrounding opinion environment affect diffusion outcomes. The experiments show that **broader propagation can be counterproductive in strongly polarized groups**, reducing rather than improving net positive influence.

[4] **Jialing Dai**, Peikun Ni, Ke Su, Jianming Zhu, Weili Wu, Ding-Zhu Du.
"Influence Maximization in Online Social Networks: Models, Algorithms, and Trustworthy Decision-Making."
Expert Systems with Applications (**CAS Zone 1 (Top), JCR Q1, IF 9.4**), article 134131, 2026. Review article. [DOI ↗]

Co-developed the **survey framework and taxonomy**, synthesized the literature, and drafted and revised the manuscript. Organized influence maximization around the relationships among network representations, behavioral assumptions, diffusion models, optimization objectives, algorithmic methods, and decision requirements. Compared greedy and lazy-greedy algorithms, sampling and sketching, mathematical programming, non-submodular search, learning-based seed selection, and adaptive policies. Examined how group effects, competition, changing networks, and objectives beyond node coverage alter the applicability of existing methods. Framed **trustworthy influence decision-making** in terms of robustness, fairness, privacy, harm control, causality, accountability, and reproducible evaluation. The resulting synthesis clarifies modeling assumptions, **theoretical guarantees, evaluation evidence, and application boundaries**, providing a methodological framework for research on network and platform decision systems.

[5] **Jialing Dai**, Yefeng Sun, Ding-Zhu Du, Weili Wu, Jianming Zhu.
"Orthant-Submodular Maximization: Polynomial-Time Approximation and Sharp Asymptotics."
SIAM Journal on Computing **(SICOMP, CCF A)**, 2026. **Under Review.**

Studied discrete allocation decisions in which each resource receives one label or remains unused and additional actions can reduce total value, modeled by nonnegative orthant-submodular functions without cross-label monotonicity. Developed dynamic-overlap search, a **uniform polynomial-time randomized algorithm** that combines candidate-dependent comparison paths with joint gain–return estimates to derive separation inequalities valid at an unknown optimum, using only value queries at feasible assignments. Established **first-order matching between algorithmic guarantees and information-theoretic bounds**. For both polynomial-time and polynomial-query algorithms, derived the optimal-ratio expansion $\beta_G + c_+/k + O(k^{-2})$, where k is the number of labels, $\beta_G \approx 0.15749$ and $c_+ \approx 0.11295$. The information upper bound holds even when the set of resources used by an optimum is given in advance. Also

characterized the **exact two-label ratio of 1/4** and developed a support-sensitive three-label algorithm attaining $V_{\text{supp}} - \varepsilon$, where $V_{\text{supp}} \approx 0.18225$, using **at most 3n value queries** for n resources and any prescribed accuracy loss $\varepsilon > 0$.

[6] **Jialing Dai**, Yefeng Sun, Ding-Zhu Du, Ke Su, Weili Wu, Jianming Zhu.
"Refined Label Comparisons for Constrained Non-Monotone k-Submodular Maximization."
International Conference on Learning Representations (ICLR, **CCF A, ICORE A***), 2027.* **Under Review.**
[OpenReview ↗]

Develops **randomized greedy and continuous algorithms** for constrained non-monotone k-submodular maximization and, more generally, full-domain orthant-submodular objectives with r-wise monotonicity. Introduces Random-root, a locally minimax randomized greedy method with a **closed-form approximation guarantee under p-system constraints**. Under matroid constraints, exploits label-specific replacement losses to improve the k-submodular approximation coefficient from $\sqrt{2} - 1 (\approx 0.4142)$ to $\alpha_2 - \tau$, where $\boldsymbol{\alpha_2 \approx 0.4218}$ and $\tau > 0$ is an arbitrarily small accuracy loss. Characterizes how additional labels strengthen the guarantees, including $0.4366 - \tau$ for three to five labels and a uniform $0.4259 - \tau$ for all $k \geq 3$. Establishes finite randomized implementations, with $\boldsymbol{O(nk\rho)}$ **value queries** for the k-submodular baseline at fixed accuracy, where n is the ground-set size and ρ the matroid rank. Extends the guarantees to real-valued objectives measured relative to the empty assignment.

[7] **Jialing Dai**, Jianming Zhu, Guoqing Wang, Jun Huang.
"Net Positive Information Diffusion Activity Maximizing in Signed Online Social Networks."
Chinese Journal of Management Science (**CSSCI, CSCD, PKU Core Journals**) 33(3): 139–150, 2025. [In Chinese.] [DOI ↗]

Formulated net-positive information activity maximization under a Signed Independent Cascade model, **jointly selecting seed nodes and their initial positive or negative states**. Measured diffusion benefits through edge-level positive and negative interactions, rather than the number of reached users alone. Established NP-hardness and #P-hardness and analyzed the interaction effects that make the original objective non-monotone and neither submodular nor supermodular. Constructed a **path-based submodular coverage surrogate** using signed propagation paths and approximate path gains, then developed a **maximum-coverage greedy algorithm** with CELF acceleration. Experiments on real signed networks distinguish reach from beneficial activity and show why a strategy that **reaches more users need not produce better net-positive word-of-mouth outcomes**.

OTHER JOURNAL PAPERS

[8] Yefeng Sun, **Jialing Dai**, Liang Gong, et al. "CaMT: Class-aware Multi-level Training Distillation for Lightweight Semantic Segmentation in Apple Orchard." *Artificial Intelligence in Agriculture* (**CAS Zone 1 (Top), JCR Q1, IF 16.1**), 2026. [DOI ↗]

[9] Yefeng Sun, **Jialing Dai**, Liang Gong, et al. "PROMO-Orchard: A Robust Semantic-Enhanced Framework for Dual-Robot Simultaneous Localization and Mapping Merging in Orchard." *Journal of Field Robotics* (**CAA A**) 43(7): 4880–4903, 2026. [DOI ↗]
Featured on the cover of the October 2026 issue.

[10] Yefeng Sun, Wei Zhang, **Jialing Dai**, et al. "A Robot-Edge-Cloud Orchestration Framework for Heterogeneous Multi-Robot Systems in Greenhouse Production." *IEEE Robotics and Automation Letters* (RA-L), 2026. **Under Review**.

OTHER CONFERENCE PAPERS

[11] Ke Su, **Jialing Dai (corresponding author)**, Weili Wu. "Scenario-Robust Checkpoint Cover for AI-Generated Content Provenance." *International Conference on Algorithmic Aspects in Information and Management (AAIM)*, 2026. Accepted / Springer LNCS (**EI Compendex** series).

Contributed to **robust checkpoint-placement modeling** for AI-generated content provenance under uncertain source and propagation scenarios, and to a **reverse-sampling-based solution** with bicriteria and high-probability guarantees.

[12] Yisheng Zhou, **Jialing Dai**, Dongyu Mao, Ke Su, Weili Wu. "Modeling Organizational Performance through Competitive Resource Diffusion." *International Conference on Algorithmic Aspects in Information and Management (AAIM)*, 2026. Accepted / Springer LNCS (**EI Compendex** series).

[13] Yefeng Sun, Liang Gong, **Jialing Dai**, Bishu Gao, et al. "S²BEV: Lightweight, Robust, and Precise SLAM-Oriented Segmentation Bird Eye's View Mapping Approach." *IEEE International Conference on Robotics and Automation (ICRA,* **CCF B, ICORE A***), Atlanta, Georgia, USA, 2025, pp. 13160–13165. [DOI ↗]

RESEARCH FUNDING AND PROJECTS

CAST Young Elite Scientists Sponsorship Program — Doctoral Student Special Program Jan. 2025 – Dec. 2026
Awardee, inaugural 2024 cohort
China Association for Science and Technology (CAST)
Research funding for opinion propagation, group-aware influence optimization, and academic exchange.

2025 International Young Partners Program Dec. 2025 – May 2026
Proposal and Project Lead | Completed
Foundation for the Development of Science and Technology in China (FDSTC)
Led academic exchange with Dr. Hongmin (William) Du, University of Massachusetts Boston, on large language models, cybersecurity information disclosure, and data-driven risk analytics.

Risk and Safety Analytics for Power-Grid Construction Oct. 2022 – Dec. 2022
Student Team Lead, under faculty supervision
State Grid Henan Electric Power Research Institute
Led a six-member student team to build a 1,498-case dataset and hierarchical safety taxonomy. Conducted association-rule and clustering analyses and developed construction-safety management recommendations.

Hazard-Source Tracing with Fault-Tree Modeling May 2022 – Apr. 2023
Research Contributor, under faculty supervision
Contributed to fault-tree analysis, hazard-source tracing, and case studies; wrote major report sections.

TEACHING EXPERIENCE

Teaching Assistant 2021–2023
University of Chinese Academy of Sciences, School of Engineering Science
Courses: Mathematical Methods and Their Applications in Management; Accounting and Cost Management; Human Factors Engineering. Supported tutorials, office hours, and student projects.

Peer Instructor 2018–2020
Beijing University of Chemical Technology, Peer Academic Development Center

ACADEMIC SERVICE

- **Program Committee Member:** AAIM 2026.
- **Journal Reviewer:** IEEE Transactions on Networking; Expert Systems with Applications (ESWA); Engineering Applications of Artificial Intelligence (EAAI); IEEE Transactions on Computational Social Systems; World Wide Web.
- **Research Seminar Coordination (UCAS, 2020–2025):** Paper selection, seminar moderation, and methodological feedback.

HONORS AND AWARDS

CAST Young Elite Scientists Sponsorship Program — Doctoral Student Special Program 2024

National Scholarship for Graduate Students 2022

National Scholarship for Undergraduate Students 2018, 2019

Outstanding Young Scholar Paper Award, Second Prize 2021
International Symposium on Emergency Management

Outstanding Graduate of Beijing 2020

Last updated: Nov. 9, 2026

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